

LEXAN™ FR Resin 945U - Asia

Polycarbonate
SABIC

PROSPECTOR®

www.ulprospector.com

Technical Data

Product Description

LEXAN™ 945U resin is a 10 MFR polycarbonate, MVR of 10. UV stabilized. Mold release. Non-chlorinated, non-brominated flame retardant, UL94 V0 rated. UL746C f1 rated. Available in opaque colors.

General

Material Status	• Commercial: Active		
UL Yellow Card ¹	• E207780-102636968 • E207780-228407 • E207780-565275 • E207780-228404		
Search for UL Yellow Card	• SABIC		
Availability	• Asia Pacific		
Uses	• Appliances • Construction Applications • Electrical Parts	• Electrical/Electronic Applications • Electronic Displays • Lawn and Garden Equipment	• Lighting Applications • Medical/Healthcare Applications • Outdoor Applications
Also Available In	• Europe	• Latin America	• North America

Physical	Nominal Value Unit	Test Method
Density / Specific Gravity	1.19 g/cm³	ASTM D792
Melt Mass-Flow Rate (MFR) (300°C/1.2 kg)	10 g/10 min	ASTM D1238
Molding Shrinkage - Flow (3.20 mm)	0.60 to 0.80 %	Internal Method
Outdoor Suitability	f1	UL 746C

Mechanical	Nominal Value Unit	Test Method
Tensile Modulus ³	2270 MPa	ASTM D638
Tensile Strength ⁴		ASTM D638
Yield	62.0 MPa	
Break	65.0 MPa	
Tensile Elongation ⁴		ASTM D638
Yield	6.0 %	
Break	130 %	
Flexural Modulus ⁵ (50.0 mm Span)	2340 MPa	ASTM D790
Flexural Strength ⁵ (Yield, 50.0 mm Span)	101 MPa	ASTM D790

Impact	Nominal Value Unit	Test Method
Notched Izod Impact (23°C)	800 J/m	ASTM D256
Instrumented Dart Impact 23°C, Total Energy	73.0 J	ASTM D3763



Thermal	Nominal Value Unit	Test Method
Deflection Temperature Under Load		ASTM D648
0.45 MPa, Unannealed, 3.20 mm	137 °C	
1.8 MPa, Unannealed, 3.20 mm	126 °C	
Vicat Softening Temperature	143 °C	ASTM D1525 ⁶
CLTE		ASTM E831
Flow : -40 to 40°C	6.7E-5 cm/cm/°C	
Transverse : -40 to 40°C	6.7E-5 cm/cm/°C	
RTI Elec	130 °C	UL 746B
RTI Imp	120 °C	UL 746B
RTI Str	130 °C	UL 746B
Electrical	Nominal Value Unit	Test Method
Arc Resistance ⁷	PLC 7	ASTM D495
Comparative Tracking Index (CTI)	PLC 3	UL 746A
High Amp Arc Ignition (HAI) ⁸	PLC 2	UL 746A
High Voltage Arc Resistance to Ignition (HVAR)	PLC 3	UL 746A
Hot-wire Ignition (HWI)	PLC 2	UL 746A
Flammability	Nominal Value Unit	Test Method
Flame Rating (1.5 mm)	V-0	UL 94
Glow Wire Flammability Index (1.0 mm)	960 °C	IEC 60695-2-12
Glow Wire Ignition Temperature (1.5 mm)	800 °C	IEC 60695-2-13
Injection	Nominal Value Unit	
Drying Temperature	120 °C	
Drying Time	3.0 to 4.0 hr	
Suggested Max Moisture	0.020 %	
Suggested Shot Size	40 to 60 %	
Rear Temperature	270 to 295 °C	
Middle Temperature	280 to 305 °C	
Front Temperature	295 to 315 °C	
Nozzle Temperature	290 to 310 °C	
Processing (Melt) Temp	295 to 315 °C	
Mold Temperature	70 to 95 °C	
Back Pressure	0.300 to 0.700 MPa	
Screw Speed	40 to 70 rpm	
Vent Depth	0.025 to 0.076 mm	
Injection Notes		

- Drying Time (Cumulative): 48 hr



Notes

¹ A UL Yellow Card contains UL-verified flammability and electrical characteristics. UL Prospector continually works to link Yellow Cards to individual plastic materials in Prospector, however this list may not include all of the appropriate links. It is important that you verify the association between these Yellow Cards and the plastic material found in Prospector. For a complete listing of Yellow Cards, visit the UL Yellow Card Search.

² Typical properties: these are not to be construed as specifications.

³ 50 mm/min

⁴ Type I, 50 mm/min

⁵ 1.3 mm/min

⁶ Rate A (50°C/h), Loading 2 (50 N)

⁷ Tungsten Electrode

⁸ Surface



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Where to Buy

Supplier

SABIC

Web: <http://www.sabic.com/>

Distributor

Please contact the supplier to find a distributor for LEXAN™ FR Resin 945U - Asia

